NetEvidence AI

Evidence-Based IP Network-Type Classification

Classify with evidence. Explain uncertainty. Make every result auditable.

Team Code: A3-PS010-TC192 • **Team: Evolve**

AIORI Hackathon 3 • PS-010

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Problem Description

THE CHALLENGE

An IP address alone does not reliably reveal the type of network or user connection behind it.

- One operator or ASN can serve multiple network environments.
- Evidence can be missing, stale, or contradictory.
- A single category label may hide uncertainty.
- Results need to be explainable and testable.

INPUT GAP
An IP address is only a starting point.

EVIDENCE GAP
Signals may disagree or be incomplete.

TRUST GAP
Confidence can be hard to interpret.

EVALUATION GAP
Claims need comparison on labelled data.

Our focus: transparent decisions, not just labels.

Background

IP ADDRESS
Identifies an endpoint or interface; it does not inherently encode its service type.

ASN & BGP
Help identify the network announcing an IP prefix and its routing context.

DNS & METADATA
Can provide additional clues, but may be absent or non-specific.

NETWORK TYPES TO CLASSIFY

Residential	Home ISP / broadband
Mobile	Mobile operator network
Enterprise / Campus	Organizational network
Hosting / Datacenter	Cloud or server infrastructure
Unknown	Insufficient or conflicting evidence

Design principle

Treat every data source as evidence—not ground truth. Track provenance, freshness, missing fields and conflicts before making a classification.

Literature & RFC Review

AREA	EXAMPLE	WHAT IT PROVIDES	OUR USE
IP intelligence APIs	IPinfo / Blackbox	Provide IP intelligence and, depending on service, network-type or hosting classification.	Baseline comparison
Routing data	RIPEstat / BGP	Provides routing, prefix and ASN-related evidence; not a complete classifier by itself.	Evidence source
BGP standard	RFC 4271	Defines Border Gateway Protocol used to exchange reachability information.	Routing context
DNS standards	RFC 1034 / RFC 1035	Describe the Domain Name System; reverse DNS can be one supporting signal.	Metadata context
ML evaluation	Scikit-learn calibration docs	Guidance for evaluating probability estimates and calibration.	Confidence evaluation

Proposed Solution

01
Validate
Check IP format and normalize input.

02
Collect
Fetch permitted ASN, BGP, DNS and metadata signals.

03
Assess
Track source, timestamp, freshness and missing fields.

04
Classify
Run a rules baseline; add ML only with labelled data.

05
Challenge
Detect conflicts; calculate calibrated confidence and evidence quality.

06
Explain
Show category or abstention, evidence graph and audit report.

Output: likely network type • calibrated confidence • evidence quality • source trace • conflicts • unknown when insufficient

Optimization Proposed

CAPABILITY	BASIC / EXISTING PATTERN	NETEVIDENCE AI ADDITION
Evidence	Single API or separate lookups	Combined view of source-tagged signals
Conflicts	May be hidden or handled differently	Explicit agreement / conflict summary
Confidence	Meaning varies by product	Calibrated confidence plus separate quality score
Unknown cases	Output depends on provider	Threshold-based abstention with reason
History	Varies by product	Saved snapshots and evidence-change alerts
Evaluation	Provider-specific methodology	Reproducible benchmark and error analysis

These are proposed improvements to test—not claims that every existing tool lacks these capabilities.

Solution Architecture & Design

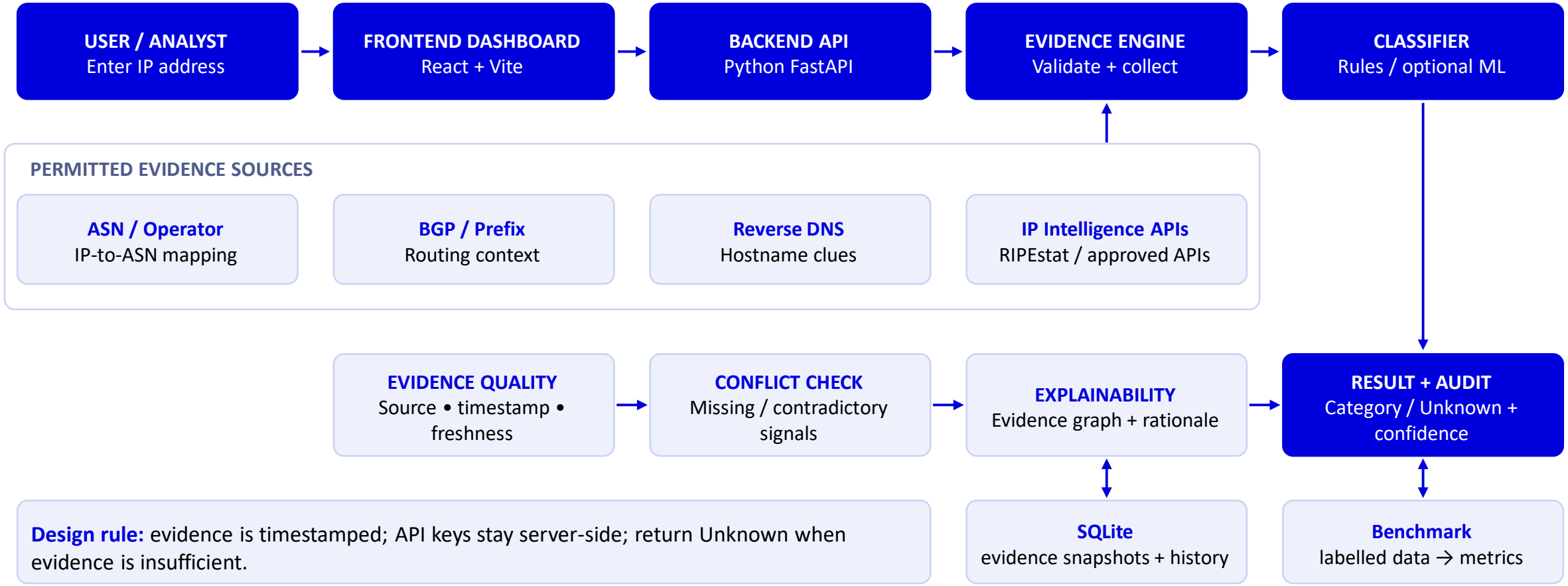


Figure: System Architecture Diagram

Data Flow Diagram

DFD LEVEL 1 | IP input → evidence → classification → explainable report

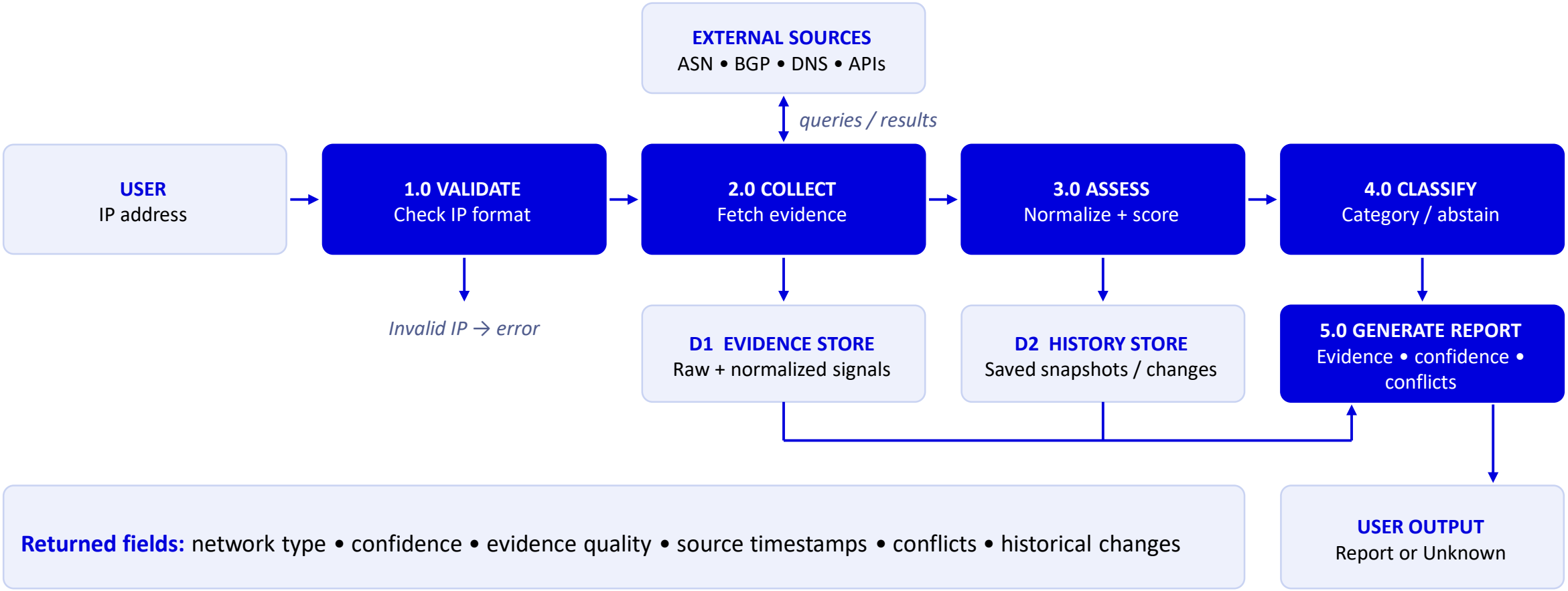


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Week 1	**Requirements & data access**	Confirm PS-010 requirements, allowed sources, labels and evaluation protocol.
Week 2	**Core prototype**	IP validation, evidence collector, source metadata and baseline rules.
Week 3	**Explainability**	Conflict detection, confidence calibration, quality scoring and evidence graph.
Week 4	**History & UI**	Save snapshots, compare evidence, build dashboard and audit report.
Week 5	**Benchmark & harden**	Evaluate on held-out prefixes, inspect errors, test missing-source behavior.
Week 6	**Demo & submission**	Deploy, document limitations, record demo video and finalize repository.

Timeline is a proposed six-week plan; adapt to the official AIORI schedule and team capacity.

Bibliography

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Thank You

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Questions & Discussion

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